

TX4 final modal-scope correction report

Finite audit of the frozen IEEE-39 phasor-domain blind-prediction campaign

FINAL CASE: A

Executive verdict

This correction audit fixes the closure sign convention, replaces invalid normalized local-factor evidence with physical $I+M_{ji}$ factors, and separates complete-spectrum V9 truth from the targeted 0.3-1.5 Hz task. The V4 collective mechanism remains supported at P4. The targeted EM task is nearly exact, but the global V9 task remains limited and the archived FC10/default table is not same-policy evidence.

Provenance

Branch: research/tx4-final-modal-scope-audit. Starting commit: 2cbb860eb4ba053edee99c1a040c8024f3f6828e. Parent freeze: 69f200dfe1dfb74cc4f678ad25a0c3b1751d62a6. No push. No radius, repair, planner, weak-node, weak-link, IEEE-68, EMT, or new-controller campaign was run.

Key headline

Q_H approaches -1, contextual return approaches +1, and physical local factors remain regular while the collective factor closes. The IAS poster is ready with corrected scope labels. The six-page paper and TPWRS submission are not ready.

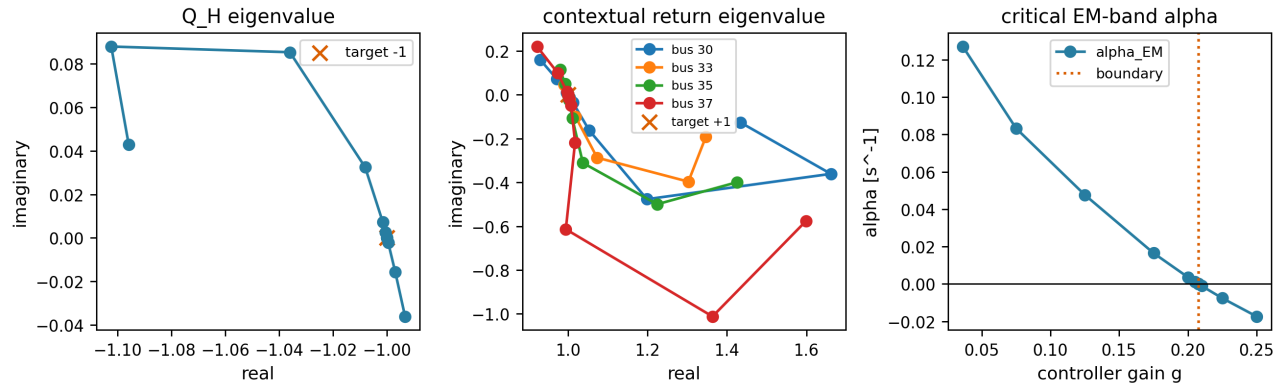
1. Sign correction and physical local factors

The exact identity is $\det(I+Q_H)=\det(I+Q_{RR}) \det(I-R_i|R)$. Therefore the collective closure condition is -1 in $\sigma(Q_H)$, while the contextual return condition is +1 in $\sigma(R_i|R)$. A previous handoff incorrectly gave both objects the same sign.

bus	abs(Q+1)	abs(return-1)	Schur residual
30	3.510e-08	1.727e-07	1.451e-16
33	3.510e-08	1.057e-07	1.303e-16
35	3.510e-08	1.118e-07	3.626e-16
37	3.510e-08	2.281e-07	1.261e-16

Sign trajectories

Correct signs: Q_H reaches -1; return reaches +1

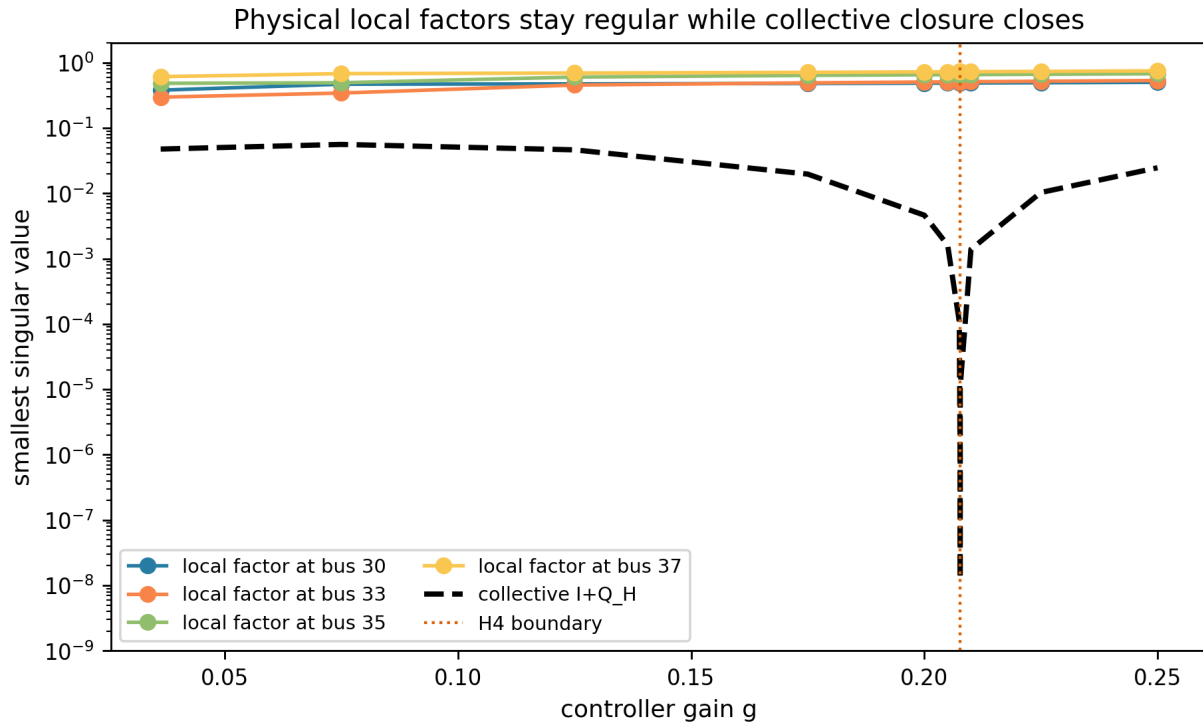


The physical audit uses $M=D K$ before normalization. The minimum physical local singular value over the frozen gain sweep is 0.2973; the collective singular value reaches $1.26e-8$. The minimum proper-subset collective factor at the boundary frequency is 0.2945.

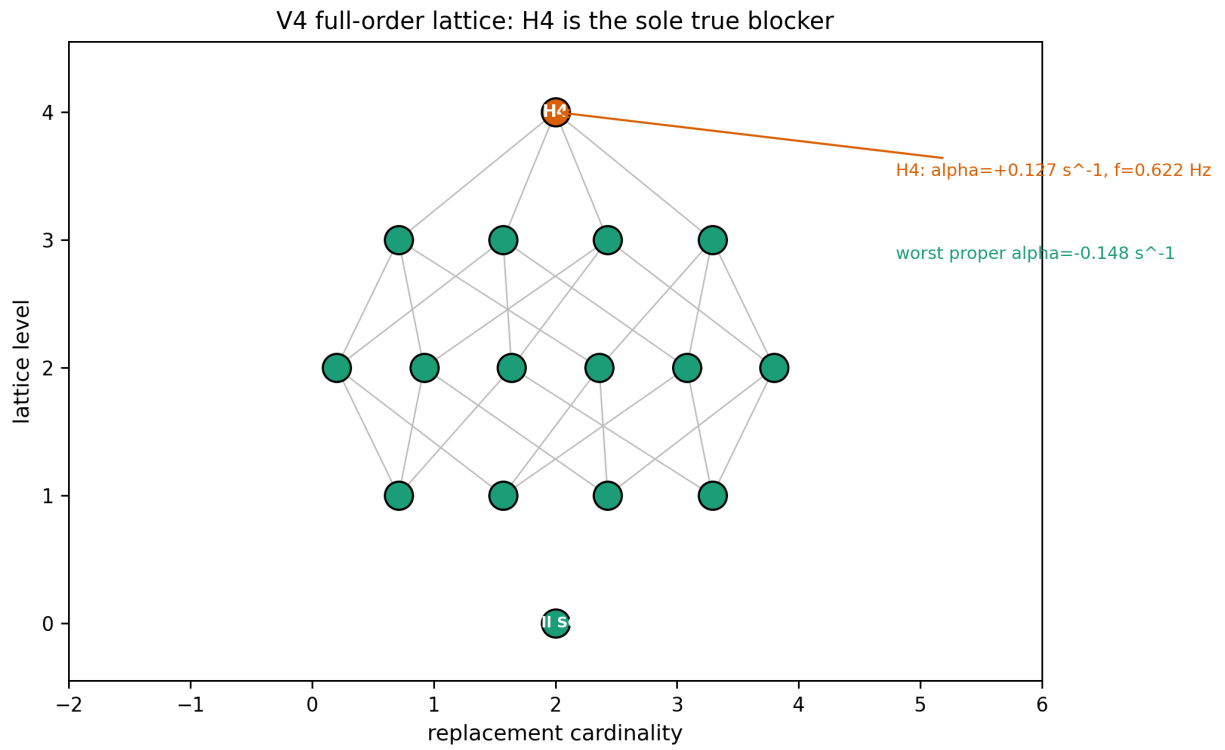
2. V4 finite blocker and collective interpretation

The V4 census is unchanged: 15 proper portfolios are stable and $H4=\{30,33,35,37\}$ is unstable at P4. H4 has $\alpha=+0.1270065 \text{ s}^{-1}$ at 0.6222797 Hz. Thus H4 is the sole true blocker in this finite V4 census, not a universal blocker for arbitrary systems.

Physical local versus collective singular values



V4 Hasse lattice



3. V9 modal-scope truth

All 512 V9 portfolios were reclassified under the same frozen P4 policy as the predictor. The complete transverse spectrum and the 0.3-1.5 Hz band are reported separately.

task	n	stable	unstable	correct	false-safe	false-unstable
global spectrum	512	332	180	402/512	110	0
0.3-1.5 Hz band	512	442	70	511/512	1	0

The archived 395/512 and 116 false-safe values came from an FC10/default policy path and are retained only as historical frozen data. They are not same-policy P4 evidence. The corrected same-policy global comparison has 110 false-safe cases and 0 false-unstable cases.

False-safe taxonomy

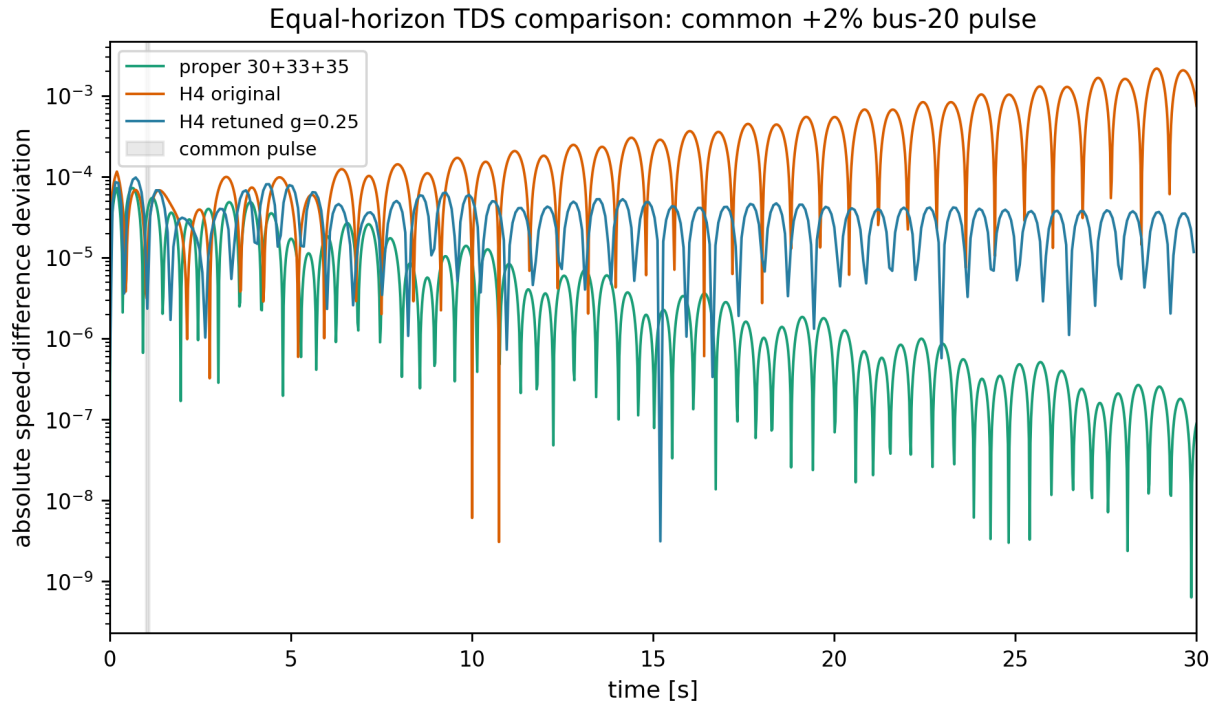
global false-safe class	count
APERIODIC_REAL	110
SLOW_OUTSIDE_BAND	0
TARGET_EM_BAND	0
FAST_OSCILLATORY	0

The one targeted EM false-safe is a separate set: 30+31+32+33+34+35+36+37 has $\alpha_{EM}=+0.04835 \text{ s}^{-1}$ at 0.7715 Hz, while its global critical mode is aperiodic. There are no eligible slow false-safe cases, so optional slow-case TDS was skipped.

4. P7 scope and nonlinear traces

P7 is SUPPLEMENT ONLY. The first member, 30+31+32+34+36+38, has global $\alpha=+422.8355 \text{ s}^{-1}$ at zero frequency but $\alpha_{EM}=-0.5313 \text{ s}^{-1}$ at 1.3774 Hz. The comparison member is stable at $\alpha=-0.1511 \text{ s}^{-1}$ and 1.3203 Hz. This is not a clean test of the validated 0.62 Hz mechanism.

Equal-horizon TDS figure



The equal-horizon TDS panel reuses the three existing phasor-domain traces over 0-30 s. It is not EMT validation and no new slow-case TDS was run.

5. Reviewer attack and claim hierarchy

Reviewer 1: novelty 5/10, correctness 8/10, model adequacy 5/10, evidence 8/10, usefulness 7/10, IAS impact 7/10, TPWRS readiness 3/10. The algebra is classical; the useful contribution is the finite blind mechanism packaging with explicit scope limits.

Reviewer 2: novelty 5/10, correctness 8/10, model adequacy 4/10, evidence 7/10, usefulness 6/10, IAS impact 6/10, TPWRS readiness 2/10. The mode is interpretable for the frozen model, but credibility is limited by one custom phasor-domain GFL, no EMT, and no second-model check.

Allowed headline

In the frozen TX4 IEEE-39 phasor-domain model at P4, the V4 H4 boundary is a collective closure loss: physical local factors remain regular while Q_H reaches -1 and its contextual return reaches +1. The targeted 0.3-1.5 Hz V9 task is nearly exact, but the complete-spectrum V9 task contains 110 same-policy aperiodic false-safe cases.

Forbidden: universal local regularity, universal V9 safety, probabilistic generalization, a new theorem, EMT validation, six-page-paper readiness, or TPWRS readiness.

6. Final decision

IAS poster redesign: YES. Use the corrected signs, physical local-factor figure, V4 lattice, and explicit global-versus-EM scope. Remove the old P7 main panel. Keep P7 in the supplement with its aperiodic/global qualification.

Six-page paper ready: NO. TPWRS ready: NO. Recommendation: NARROW. Proceed with the corrected IAS poster and external ChatGPT review of the ZIP. Defer broader journal claims until a second documented dynamic model, more operating points, controller-limit validation, and independent nonlinear validation are available.

Final answers

question	answer
true V4 blocker	$H_0=\{H_4\}$ in the finite V4 census
Q / return signs	$Q_H \rightarrow -1$; return $\rightarrow +1$
V9 global	402/512 correct; 110 false-safe, all aperiodic
V9 EM band	511/512 correct; 1 false-safe, 0 false-unstable
P7	SUPPLEMENT ONLY
TDS	equal-horizon replot only; no slow cases eligible
poster / paper / TPWRS	YES / NO / NO